

WIA2001 DATABASE DRAFTING GROUP PROJECT

Introduction

(later just explain your part above yours)

Business Rules

System objectives and scope

User requirements

Entities and attributes

ER diagram

Normalization process

Implementable ERD

Sample SQL statements

Screenshots of tables from developed system

Discussion

1. Write a detailed and accurate description of business operations and objectives
2. Develop an extensive and precisely written business rules based on the description of business operations
3. List out some of the requirements from the user
4. Define the entities, attributes, relationships, cardinalities and constraints for each module defined in step 3. Use the business rules develop in step 2 as the source for your component definitions
5. Create ERD segments to model the modules you define in step 4. Remember that each of the ERD's entities will be implemented through a database table.
6. Perform the appropriate normalization checks.
7. If necessary, modify the ERD segments you developed in step 5 to reflect the normalization findings in step 6.
8. The GROUP LEADER needs to submit the power point presentation though designated link on SPECTRUM and save it as GT(Your tutorial group number)-GroupNumber.pdf

Project 1 Presentation (10 minutes including Q & A)

The presentation slides should contain the following information :

- 1. Title (e.g. Database Management System for ABC Inc) and List of Group Members (with Metrics No)**
- 2. Presentation Outlines**
- 3. Introduction to the organization of choice (azka & win10)**
- 4. Issues in managing data (azka)**
- 5. Business Rules (azka win10)**
- 6. Entity Relationship Diagram (wendy)**
- 7. [Normalization \(yanjie\)](#)**
- 8. Challenges faced during the data collection process (win10 azka)**
- 9. Proof of the interview / data collection (screenshot of online meeting, picture of the organization with the owner etc)**

Interview Script

Interview Script

General Questions

1. Could you start by introducing yourself and sharing a bit about your role within the Rockstar Academy?
2. Could you provide an overview of your Rockstar Academy's purpose, including its vision, mission, and key objectives? What core values drive your business?
3. Could you describe your company's primary business operations? What are the main activities or services that support your mission and objectives?
4. Who are the main users of your system?
 - a. Is it mostly front desk staff, coaches, or managers who use it, and what do they each need to do?
5. Are there any special requirements or permissions when working with kids and teens?
 - a. For example, do you have specific rules for signing up minors, or do parents need to provide any approvals?
6. We'd love to learn more about your company's business rules. Could you explain the different entities involved, their relationships, and roles within your operations? For instance, who are your primary stakeholders, such as customers, trainees, or partners, and do you offer any specific membership or service plans?
7. What kinds of user requirements do you have for managing data? For example, do you need functionalities to maintain, track, or search for information about specific entities within your system?

ERD Questions

1. What key information do you track in your inventory system?
 - a. For example, details about members, coaches, classes, or equipment?
2. What details do you store for each member, as well as for each coach and class?
 - a. This could include things like names, contact information, class times, and schedules.
3. How are members, coaches, and classes connected in the system?
 - a. For instance, are kids assigned to specific coaches, or are they free to attend different classes?
4. Are there any special rules around the data you collect for members?
 - a. For example, age requirements for certain activities or maximum class sizes?

Specific Management Questions

1. How do you manage student sign-ups and class schedules across different fitness programs?

2. Do you use any tools or software for managing schedules, attendance, or tracking student progress? Are there any challenges with the current system?
3. How often do you check student attendance and progress? How do you handle frequent absences?
4. What information do you keep for each class (like class name, type, schedule, age group)?
5. How do you decide when to add more spots in a class or create new sessions?
6. How is attendance recorded, especially for make-up sessions or missed classes? Is it updated right away?
7. Are there any restrictions for classes, like certain age or skill levels? How do you keep track of this?
8. How do you handle discrepancies in attendance, like missed sessions?
9. What kinds of reports or summaries would be helpful for managing classes and tracking student engagement?
10. What's the average sign-up rate for different programs? Are some classes more popular?
 - a. Gymnast
 - b. Sport
 - c. Basketball
 - d. Martial arts
 - e. Baby jam
 - f. Ballet
11. How many students and classes are currently active at the academy?
12. Who makes the decisions about scheduling, class setup, and student progress tracking?

Meeting Minutes (IV)

Answers

General Questions

1. Area manager in rockstar academy, handling 6 academy
2. One stop physical education academy for kids, objectives are base for children help improve their physical and also team bonding activity
3. 100 classes per week based on their age, members can choose classes, also provide competitions, have private classes,
4. Data sources from sales team once become a member, use the data to upselling to members, how many classes attend, categorized to frequency of members, and types of membership, coaches use data to make progress reports for members at the end of each term (3 months)
5. Members will have contracts and everything under rockstar responsibility, class will have an age category, members can only attend based on their age group.
6. Business rules
 - a. Attend based on age group
 - b. Must bring their identification to ensure correct grouping
 - c. Got diff groups and times for classes to match beginner to pro levels
 - d. Certain teacher for certain gender
 - e. Equipment for beginners n equipment for each level
 - f. Got star teacher, coaches and etc for diff levels
 - g. Every coach can teach private
 - h. Only 1 membership type (VIP)
 - i. Normal membership for one academy only - only the area outside
 - j. As long as the age grp is suitable they can attend any class\
 - k. Must register for the class first as we have limitations for each class
 - l. There are maximum attendances, eg 30 students in 1 class only
 - m. Join together even though specially abled
 - n. Specially abled will be shadowed and extra cost ~ rm 1 mil rupiah for 10 session
 - o. Can only use the system during the class (coaches)
 - i. For reporting, signing in, recording
7. Different data for every division
 - a. Academic teacher - total no of students in 1 academy / class
 - b. Area - total of the academy within manager's jurisdiction
 - i. Sales no,
 - ii. Kpi
 - iii. How many priv
 - iv. How many diff types of membership

ERD Questions

1. Members

- a. Name
- b. Dob
- c. Address
- d. School
- e. parents info (name)
- f. tel no (parents) , email address(parents), How many classes they attend, what programmes they have (private session or public), mention the remaining sessions they have left, coaches and manager's data
- g. Coaches
 - i. Same as member's data
- h. Manager
 - i. Same as member's data

2. Special data

- a. Members can only register for 21 classes in the system but they can attend more
- b. Got maximum capacity for each classes stated
- c. Members that registered for classes will be removed if they do not attend for that registered class
- d. Progress report will be given as long as minimum attendance is met
 - i. 8 attendances within a month

1. Only add or close classes in the beginning of the term

- a. Changes can be amended during week 5
- b. Eg only close classes or open classes before 21st oct or during week 5
- c. Under 5 members - low attendance throughout the term (can close the class)
- d. Petition must be signed if a new class wants to be open
- e. Will be open after the end of the term
- f. They can register for the class anytime they want but
 - i. Management can only change the classes according to the restricted time
 - ii. If we want to add new classes then they must do it within the time period
- g. Uses axioma system
 - i. Can change the teachers in the class
 - ii. Class will be verified before it's over
 - 1. See how many kids attend
 - 2. Uis the teacher suitable ?
 - a. Using appropriate methods and uniforms?
 - 3. Teachers will be paid based of off these
 - a. Deduction will occur if teaching is not satisfactory

2. Progress reports

- a. Made by teachers at the end of the term (every 3 months)
- b. For each student who attended the classes (meets the minimum attendance)

- i. Automatic calculation
 - 1. Monday class - 20/30 students get report only
 - c. Report consists of
 - i. Gymnastics - has rolled, bar, jump (learnt what skill)
 - d. IT team will send the report every day (shared everyday during class)
 - i. How many private instructor used on that day
 - ii. Sales
 - iii. Attendance
 - iv. How many memberships from the beginning of the month till now
 - v. Kpi progress (how many sales made etc)
- 3. Class Id, class time, class type
- 4. Sometimes attendance are recorded manually to be put in the system later, consists of student id, teacher id, and class
- 5. Can not do replacement classes, because cannot do manually
- 6. We can add a social ability (MBTI?), student behavior in class, addition: parent can only company student who are under 5
- 7. Average sign ups different for each age group, gymnastics avg 30 students, basketball, martial arts are the classes that are usually full, baby jam is the least attended because the age groups are for baby, ballet for teens consists of avg 6 max 10.
- 8. Development manager

Presentation Info

1. Introduction

Rockstar Academy is the biggest one-stop sports and performing arts academy center for kids and teens in Asia, providing opportunities for children to have sufficient physical activities in a safe environment and to experience a rounded and balanced growth. Offering a full range of physical activities, from sports to performing arts, allowing their students to establish a healthy lifestyle from an early age, explore their talents, reach their full potentials, and develop essential qualities required to become Future Life Champions.

2. Objectives

- a. Efficient Member Management:
 - i. Streamline the member registration and enrollment process.
 - ii. Accurately track member demographics, contact information, and emergency contacts.
 - iii. Efficiently manage membership renewals and cancellations.
 - iv. Maintain accurate records of member attendance and participation in classes.
- b. Effective Class Management:
 - i. Optimize class scheduling and resource allocation.
 - ii. Facilitate easy class registration and enrollment.
 - iii. Track class attendance and generate attendance reports.
 - iv. Manage class capacity and waitlists.
 - v. Monitor class performance and make necessary adjustments.
- c. Accurate Financial Management:
 - i. Track membership fees, tuition payments, and other revenue streams.
 - ii. Generate accurate invoices and receipts.
 - iii. Manage payroll and expenses.
 - iv. Produce financial reports and statements.
- d. Improved Operational Efficiency:
 - i. Automate routine tasks, such as class scheduling and report generation.
 - ii. Reduce manual data entry and minimize errors.
 - iii. Enhance communication between staff, parents, and students.
 - iv. Provide real-time access to information for decision-making.
- e. Enhanced Decision-Making:
 - i. Analyze data to identify trends and patterns in member behavior, class performance, and financial performance.
 - ii. Generate insightful reports to support strategic planning and decision-making.
 - iii. Optimize resource allocation and staffing levels.
 - iv. Identify opportunities for improvement and innovation.
- f. Data Security and Integrity:
 - i. Protect sensitive member and employee data from unauthorized access.
 - ii. Ensure data accuracy and consistency through data validation and quality control measures.
 - iii. Implement data backup and recovery procedures to minimize data loss.
 - iv. Comply with relevant data privacy and security regulations.

3. Business operations

- a. **Class Scheduling:** Classes are scheduled based on age group, skill level, and class type.
- b. **Member Enrollment:** Members enroll in classes and purchase memberships.
- c. **Class Attendance:** Members attend classes and are tracked for attendance purposes.
- d. **Performance Tracking:** Coaches monitor student progress and provide feedback.
- e. **Data Management:** Data on membership, attendance, and performance is collected and analyzed.

4. Business Rules

- Database has **8 entities** namely, MEMBER, CLASS, COACH, MANAGER, ACADEMY, PROGRESS REPORT, EQUIPMENT and SPECIAL SUPPORT
- Entity MEMBER contains attributes MemberID, Name, Date_Of_Birth, Address, School, ParentID, Parent_Name, Parent_Phone, Parent_Email, Membership_Type, Total_Class_Attended, Classes_Registered and Remaining_Sessions
- Entity CLASS contains attributes ClassID, Class_Name, Class_Type, Age_Group, Skill_Level, Capacity, Start_Time, End_Time
- Entity COACH contains attributes CoachID, Name, Gender, Skill_Level, Contact_Info
- Entity MANAGER contains attributes ManagerID, ITTeam, ReportID, Name, Contact_Info, Academy_Managed
- Entity ACADEMY contains attributes AcademyID, Name, Location, Total_Classes, Total_Memberships
- Entity PROGRESS REPORT contains attributes ReportID, MemberID, CoachID, Skills_Learned, Attendance_Count
- Entity EQUIPMENT contains attributes EquipmentID, Name, Type, Quantity_Available
- Entity SPECIAL SUPPORT contains attributes SupportID, Type, Cost, Sessions_Available

- Each MEMBER can attend one-or-many CLASS
- Each MEMBER can have one-or-many PROGRESS REPORT
- Each CLASS can use one-or-many EQUIPMENT specific to the class levels
- Each CLASS can have zero-or-many SPECIAL SUPPORT for specially-abled members
- A COACH can teach one-or-many CLASS
- A COACH can generates one-or-many PROGRESS REPORT for members
- A MANAGER manage one-or-many ACADEMY
- A MANAGER handle one-or-many PROGRESS REPORT generated by COACH
- Each ACADEMY offers one-or-many CLASS
- Each PROGRESS REPORT that is generated for each MEMBER by one-or-many COACH

- MEMBERS can attend multiple CLASS and each CLASS can have multiple MEMBER has many-to-many relationships
- MEMBERS can be assigned to multiple COACH, and each COACH can have multiple MEMBERS has many-to-many relationships

- Each CLASS is taught by one COACH, and each COACH can teach multiple CLASS has one-to-many relationships
- MEMBER has multiple PROGRESS REPORT generated by COACH is one-to-many relationships
- CLASS use specific EQUIPMENT based on skill level has one-to-many relationships
- Each ACADEMY is managed by one MANAGER and a MANAGER can handle multiple ACADEMY has one-to-many relationships
- SPECIAL SUPPORT is provided for MEMBER in specific classes, if required has one-to-many relationships

5. User requirements

Member Management:

- Support registration and enrollment of new members.
- Store and manage member details, including demographics, emergency contacts, and membership type.
- Track member attendance and participation in classes.
- Facilitate membership renewals and cancellations.

Class Management:

- Enable creation and scheduling of classes by type, age group, skill level, and capacity.
- Allow members to register for available classes.
- Track class attendance and manage class capacity, including waitlists.
- Provide performance monitoring for classes, enabling adjustments based on attendance and feedback.

Financial Management:

- Record membership fees, payments, and other revenue sources.
- Generate accurate invoices, receipts, and financial reports.
- Track payroll and operational expenses.

Operational Efficiency:

- Automate repetitive tasks like scheduling and report generation.
- Minimize manual data entry to reduce errors.
- Enhance communication between academy staff, parents, and students.

Progress Tracking:

- Allow coaches to generate progress reports for students, including skills learned and attendance metrics.
- Provide managers access to reports for oversight and decision-making.

Special Support:

- Record and manage optional special support services, including type, cost, and session availability.

6. Entities, attributes, relationship, cardinalities and constraints**6.1 Entities and attributes**

Entity	Attributes
MEMBER	MemberID (PK)
	Name
	Date_Of_Birth
	Address
	School
	ParentID
	Parent_Name
	Parent_Phone
	Parent_Email
	Membership_Type
	Total_Class_Attended
	Class_Registered
	Remaining_Sessions

Entity	Attributes
CLASS	ClassID (PK)
	Class_Name
	Class_Type
	Age_Group

	Skill_Level
	Capacity
	Start_Time
	End_Time

Entity	Attributes
COACH	CoachID (PK)
	Name
	Gender
	Skill_Level
	Contact_Info

Entity	Attributes
MANAGER	ManagerID (PK)
	ReportID
	ITTeamID
	Name
	Contact_Info
	Academy_Managed

Entity	Attributes
ACADEMY	AcademyID (PK)
	Name
	Location
	Total_Classes

	Total_Membership
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Entity	Attributes
PROGRESS REPORT	ReportID (PK)
	MemberID (FK)
	CoachID (FK)
	Skills_Learned
	Attendance_Count

Entity	Attributes
EQUIPMENT	EquipmentID (PK)
	Name
	Type
	Quantity_Available

Entity	Attributes
IT TEAM	ITTeamID (PK)
	ReportID (FK)
	Name
	Contact_Info

	KPI_Data
	Sales

Entity	Attributes
SPECIAL SUPPORT (OPTIONAL)	SupportID (PK)
	Type
	Cost
	Session_Available

6.2 Relationship, cardinalities and constraints

6.2.1 Relationship between entities

MEMBER

- Attends multiple CLASS (Many-to-Many)
- Has PROGRESS REPORT generated (One-to-Many)

COACH

- Teaches multiple CLASS (One-to-Many)
- Generates PROGRESS REPORT for MEMBER (One-to-Many)

CLASS

- Has multiple MEMBER (Many-to-Many)
- Assigned to a COACH (One-to-Many)
- Uses specific EQUIPMENT (One-to-Many)
- Related to Special Support for specially-abled members (One-to-Many)
- Offered by Academy (One-to-Many)

MANAGER

- Manages multiple ACADEMY (One-to-Many)
- Handles multiple PROGRESS REPORT (One-to-Many)

ACADEMY

- Managed by one MANAGER (One-to-Many)
- Offers multiple CLASS (One-to-Many)

PROGRESS REPORT

- Generated by COACH (One-to-Many)

- Handled by MANAGER (One-to-Many)
- For each MEMBER (One-to-Many)

EQUIPMENT

- Assigned to CLASS based on necessity (One-to-Many)

SPECIAL SUPPORT

- Assigned to CLASS based on requirement (One-to-Many)

6.2.2 Cardinalities and constraints

MEMBER

- MEMBER and CLASS: Many-to-Many
 - A member can attend many classes, and each class can have many members.
 - A member is associated with one or more classes with a possibility of zero, and a class is associated with one or more members with a possibility of zero.

COACH

- COACH and CLASS: One-to-Many
 - A coach can teach many classes, and each class can only have one coach.
 - A coach is associated with one or more classes, and a class is associated with one coach.
- COACH and PROGRESS REPORT: One-to-Many
 - Each coach generates multiple progress reports for members.
 - A coach is associated with many progress reports, but a report is associated with exactly one coach.

CLASS

- CLASS and MEMBER: Many-to-Many
 - A class can have many members, and each member can attend multiple classes.
 - A class is associated with many members, and a member is associated with many classes.
- CLASS and COACH: Many-to-Many
 - A class can have only one coach, and a coach can teach multiple classes.
 - A class is associated with exactly one coach, and a coach is associated with one or more classes.
- CLASS and EQUIPMENT: One-to-Many
 - A class uses specific equipment based on skill level.

- A class can use many pieces of equipment, and each piece of equipment is assigned to a specific one class.
- CLASS and SPECIAL SUPPORT: Optional One-to-Many
- CLASS and ACADEMY: One-to-Many
 - Each Academy can offer many classes, but each Class belongs to exactly one Academy.
- Special needs support is optional and provided for specific classes when required.
- A class may have zero or many special needs support, and special needs support is provided for one or more classes.

MANAGER

- MANAGER and ACADEMY: One-to-Many
 - A manager manages one or more academies, and each academy has exactly one manager.
 - A manager is associated with one or more academies, and an academy is associated with exactly one manager.
- MANAGER and PROGRESS REPORT: One-to-Many
 - A manager handles multiple reports, and each report is handled by one manager.
 - A manager is associated with many reports, and each report is associated with one manager.

ACADEMY

- ACADEMY and MANAGER: One-to-Many
 - An academy is managed by exactly one manager, but a manager can manage many academies.
 - An academy is associated with one manager, and a manager is associated with many academies.
- ACADEMY and CLASS: One-to-Many
 - An academy can offer many classes, but each class is offered by exactly one academy.
 - An academy is associated with one or more classes, and a class is associated with exactly one academy.

PROGRESS REPORT

- PROGRESS REPORT and MEMBER: One-to-Many
 - Each progress report is associated with one member, and each member can have many progress reports.
 - A member is associated with one or more progress reports, and a report is associated with exactly one member.
- PROGRESS REPORT and COACH: One-to-Many

- Each progress report is associated with one coach, and each coach can generate multiple reports.
 - A coach is associated with many reports, and each report is associated with exactly one coach.
- PROGRESS REPORT and MANAGER: One-to-Many
 - Many progress reports are handled by one manager, and each manager can handle multiple progress reports.
 - A manager can handle many progress reports, and each report is handled by exactly one manager.

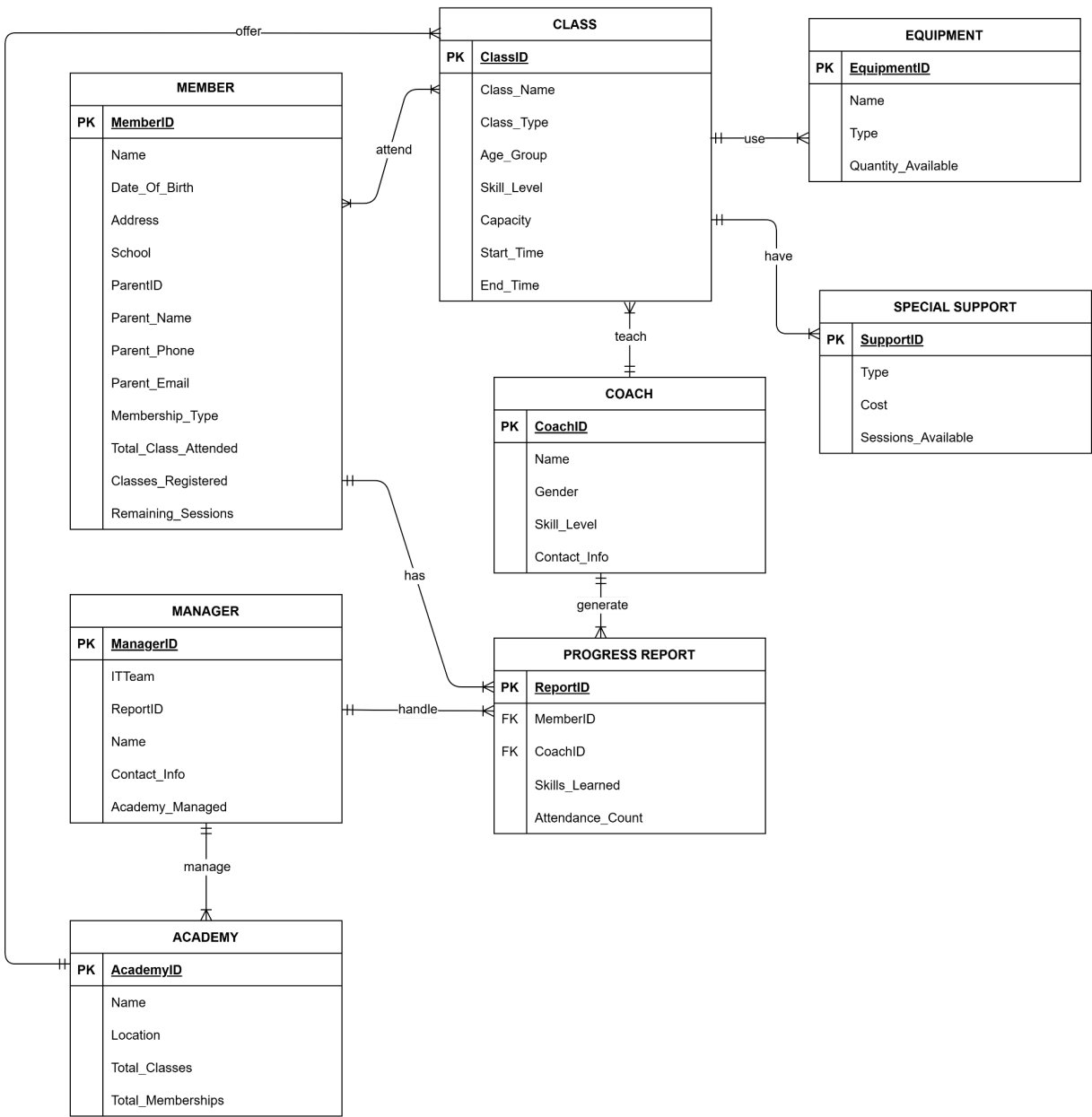
EQUIPMENT

- EQUIPMENT and CLASS: One-to-Many
 - Each class can use multiple pieces of equipment, and each piece of equipment can be used by certain classes.

SPECIAL SUPPORT

- SPECIAL SUPPORT and CLASS: One-to-Many
 - Special support can be provided for specific classes when required.
 - A class may have zero or many special supports, and each special support is associated with one class

ERD



Description of the Entities

MEMBER: Represents individuals registered with the academy, including details about their membership and sessions attended.

CLASS: Represents the classes offered, categorized by type, skill level, and other details.

COACH: Represents the coaches who conduct classes, along with their expertise and contact information.

EQUIPMENT: Represents equipment used in classes, detailing their type and availability.

SPECIAL SUPPORT: Represents additional support services available for members, including costs and sessions.

PROGRESS REPORT: Tracks the progress of members, including skills learned and attendance, generated by coaches.

MANAGER: Represents the managers overseeing academy operations and handling progress reports.

ACADEMY: Represents the academy, detailing its location and overall memberships.

Relationships Between Entities

MEMBER attends CLASS: Members register and attend classes.

COACH teaches CLASS: Each class is conducted by a coach.

CLASS uses EQUIPMENT: Classes require specific equipment.

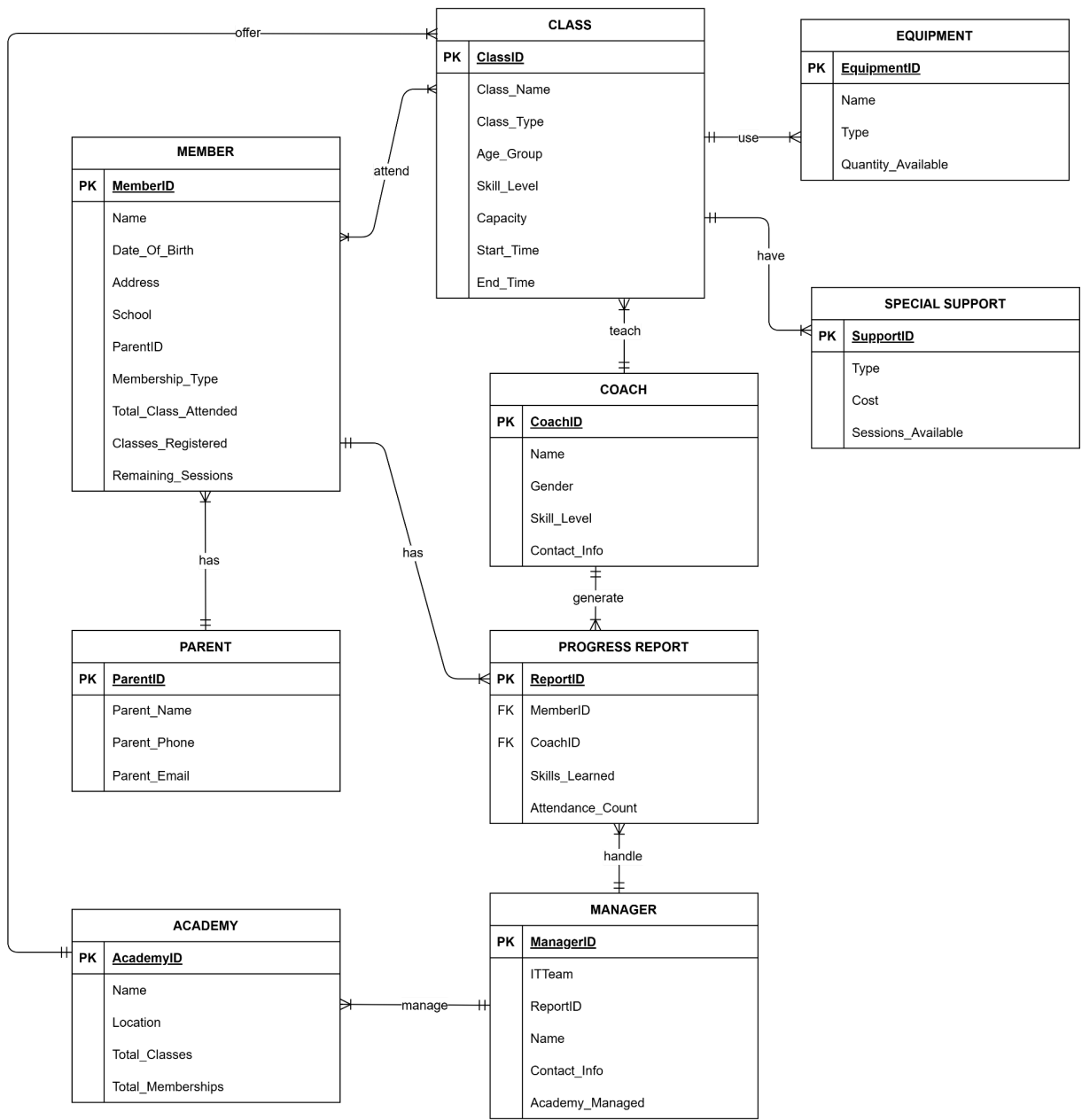
MEMBER may need SPECIAL SUPPORT: Members can utilize support services.

COACH generates PROGRESS REPORT: Coaches prepare reports tracking members' progress.

MANAGER handles PROGRESS REPORT: Managers oversee the creation and maintenance of reports.

MANAGER manages ACADEMY: Managers are responsible for specific academies.

ERD after Normalisation



Normalization Process

Normalization plays a crucial role in organizing the data structure for the Rockstar Academ's database, optimizing data storage and minimizing redundancy. In this report, we detail the normalization process for Rockstar Academy's database, transforming it from the initial Unnormalized Form (UNF) to the Third Normal Form (3NF). This process includes identifying and addressing partial and transitive dependencies while structuring the database into well-organized, efficient tables.

Unnormalised Form (UNF)

The Unnormalized Form (UNF) represents the raw data in its most disorganized state. This form contains redundancy and dependency issues, making the data difficult to work with. The tables at this stage may have repeating groups, and some attributes depend on only part of a composite primary key or other attributes.

MEMBER

MemberID	Name	Date_Of_Birth	Address	School	Parent ID	Parent _Name	Parent_Phone	Parent_Email
Membership_Type		Total_Class_Attended	Class_Registered	Remaining_Sessions				

CLASS

Class_ID	Class_Name	Class_Type	Age_Group	Skill_Level	Capacity	Start_Time	End_Time
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COACH

CoachID	Name	Gender	Skill_Level	Contact_Info
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MANAGER

ManagerID	ReportID	Name	Contact_Info	Academy_managed
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ACADEMY

AcademyID	Name	Location	Total_Classes	Total_Membership
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PROGRESS REPORT

ReportID	MemberID	CoachID	Skills_Learned	Attendance_Count
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EQUIPMENT

Equipment_ID	Equipment_Name	Equipment_Type	Quantity_Available
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SPECIAL SUPPORT

SupportID	Type	Cost	Sessions_Available
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First Normal Form (1NF)

To move from UNF to 1NF, we need to eliminate repeating groups and ensure that all attributes contain atomic (indivisible) values. Additionally, we identify the Primary Key (PK) for each table.

MEMBER

MemberID	Name	Date_Of_Birth	Address	School	Parent ID	Parent_Name	Parent_Phone	Parent_Email
Membership_Type		Total_Class_Attended	Class_Registered	Remaining_Sessions				

CLASS

Class_ID	Class_Name	Class_Type	Age_Group	Skill_Level	Capacity	Start_Time	End_Time
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COACH

CoachID	Name	Gender	Skill_Level	Contact_Info
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MANAGER

ManagerID	ReportID	Name	Contact_Info	Academy_managed
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ACADEMY

AcademyID	Name	Location	Total_Classes	Total_Membership
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PROGRESS REPORT

ReportID	MemberID	CoachID	Skills_Learned	Attendance_Count
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EQUIPMENT

Equipment_ID	Equipment_Name	Equipment_Type	Quantity_Available
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SPECIAL SUPPORT

SupportID	Type	Cost	Sessions_Available
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Second Normal Form (2NF)

In 2NF, we eliminate partial dependencies, which occur when a non-key attribute is dependent on only part of the primary key.

For the Rockstar Academy database, there are no partial dependencies present.

MEMBER

MemberID	Name	Date_Of_Birth	Address	School	Parent ID	Parent_Name	Parent_Phone	Parent_Email
Membership_Type		Total_Class_Attended	Class_Registered	Remaining_Sessions				

CLASS

Class_ID	Class_Name	Class_Type	Age_Group	Skill_Level	Capacity	Start_Time	End_Time
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COACH

CoachID	Name	Gender	Skill_Level	Contact_Info
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MANAGER

ManagerID	ReportID	Name	Contact_Info	Academy_managed
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ACADEMY

AcademyID	Name	Location	Total_Classes	Total_Membership
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PROGRESS REPORT

ReportID	MemberID	CoachID	Skills_Learned	Attendance_Count
----------	----------	---------	----------------	------------------

EQUIPMENT

Equipment_ID	Equipment_Name	Equipment_Type	Quantity_Available
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SPECIAL SUPPORT

SupportID	Type	Cost	Sessions_Available
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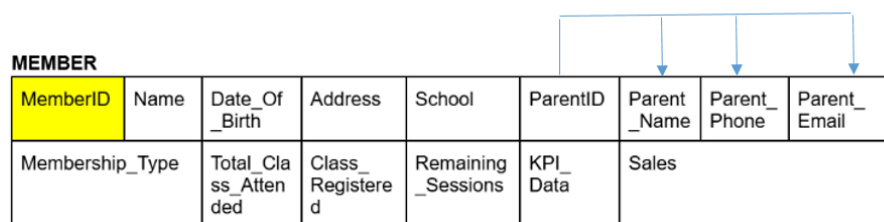
ACADEMY

AcademyID	Name	Location	Total_Classes	Total_Membership
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Third Normal Form (3NF)

In 3NF, we remove transitive dependencies, which occur when a non-key attribute depends on another non-key attribute through the primary key.

The attributes Parent_Name, Parent_Phone and Parent_Email depend on ParentID. However, ParentID is indirectly related to the primary key, MemberID. This creates a transitive dependency.



To eliminate this transitive dependency, we create a separate table called PARENT INFO. This new table will have ParentID as its primary key and will store the parent-related details, which are Parent_Name, Parent_Phone, and Parent_Email. The MEMBER table will then reference this new PARENT INFO table through a foreign key relationship, where ParentID in the MEMBER table links to the ParentID in the PARENT INFO table. This resolves the transitive dependency by moving parent details into a distinct entity, ensuring that all attributes in the MEMBER table now depend directly on the primary key MemberID.

MEMBER

MemberID	Name	Date_Of_Birth	Address	School	ParentID
Membership_Type		Total_Class_Attended	Class_Registered	Remaining_Sessions	

PARENT INFO

ParentID	Parent_Name	Parent_Phone	Parent_Email
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MEMBER

MemberID	Name	Date_Of_Birth	Address	School	ParentID
Membership_Type		Total_Class_Attended	Class_Registered	Remaining_Sessions	

PARENT INFO

ParentID	Parent_Name	Parent_Phone	Parent_Email
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CLASS

Class_ID	Class_Name	Class_Type	Age_Group	Skill_Level	Capacity	Start_Time	End_Time
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COACH

CoachID	Name	Gender	Skill_Level	Contact_Info
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MANAGER

ManagerID	ReportID	Name	Contact_Info	Academy_managed
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ACADEMY

AcademyID	Name	Location	Total_Classes	Total_Membership
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PROGRESS REPORT

ReportID	MemberID	CoachID	Skills_Learned	Attendance_Count
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EQUIPMENT

Equipment_ID	Equipment_Name	Equipment_Type	Quantity_Available
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SPECIAL SUPPORT

SupportID	Type	Cost	Sessions_Available
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Script

Intro

1. Greetings
 - a. Good afternoon
 - b. Group7
2. Intro
 - a. Rockstar academy
 - i. A sports and performing arts academy center for kids and teens in Asia and the specific branch that we had interviewed was in indonesia at the time
 - ii. This academy offers gymnastics, basketball, ballet, baby jam for many types of members that come from multiple age groups
3. Objectives
 - a. Efficient class and member management

Challenges and Proof of interview

1. Communication Barriers
 - a. Accent
2. Time conflict
 - a. Schedule
3. Interview platform
 - a. Free
4. Proof
 - a. Mr Yogi, from Indonesia ~ lasted about two hours
 - b. Wonderful session and we learned alot on the behind the scenes of how he manages many academies efficiently